

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		inert pair effect becomes more significant down the group	1			1		
		(ii)		$\text{PbO}_2 + 4\text{HCl} \rightarrow \text{PbCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$		1		1		
	(b)	(i)		1 st order with respect to hydrogen ions (1) 1 st order with respect to tin ions / 0 th order with respect to iron(III) ions (1) $\text{rate} = k[\text{H}^+][\text{Sn}^{2+}]$ (1)		1	1 1	 3	1 1	
		(ii)		$k = \text{rate} / [\text{H}^+][\text{Sn}^{2+}]$ (1) ecf from rate equation given in part (i) $k = 0.14$ (1)		2		2	2	
		(iii)		not correct – must give a sensible reason (1) as the rate determining step should include one hydrogen ion / one tin ion / no iron(III) ion (1) ecf from rate equation given in part (i)			1 1	 2		
	(c)	(i)		removing samples from the reaction mixture at set intervals and then stopping the reaction / adding large volume of water	1			1		1
		(ii)		reacting ratio $5\text{Fe}^{2+} : 1\text{MnO}_4^-$ concentration = $0.0558 \text{ mol dm}^{-3}$		2		2	2	
				Question 10 total	2	6	4	12	6	1